College	**RV College of Engineering®, Bengaluru**		
Department & Program	**Computer Science and Engineering - M.Tech in Computer Science and Engineering**		
Course:	**Major Project**	**Course Code**	**MCE442P**
Student Name	**VEERASAGAR S S**	**USN**	**1RV24SCS17**
Project Title	**ACTIS: An Agentic Collaborative Threat Intelligence System using Trust-Aware Federated Learning**		
Working Domain	**Federated IDS threat agent**		
Undertaken at	**RVCE**		
Internal Guide	**Dr. Rajashree Shettar** **Dept of CSE, RVCE**	**External Guide**	**NA**

Abstract :

As cyber threats rapidly scale in sophistication, isolated organizational defenses are consistently outpaced by zero-day attacks. While sharing network telemetry across organizations can enable a powerful collective defense, strict data privacy regulations (e.g., GDPR) and the severe risk of exposing Personally Identifiable Information (PII) prohibit the centralized pooling of raw network traffic.

To counter the statistical heterogeneity of non-IID data distributions, the system implements a FedProx-based localized training regimen coupled with a novel Trust-Aware FedAvg algorithm, which dynamically penalizes underperforming or compromised nodes. Furthermore, ACTIS introduces an LLM-driven Agentic Privacy Engine that autonomously maps network anomalies to MITRE ATTACK tactics while guaranteeing zero PII leakage through strict regex-bound sanitization. The sanitized threat intelligence is converted into semantic embeddings secured by an Privacy layer and stored in a central vector database. ACTIS achieved 98.82% federated accuracy, 0.962 ROC-AUC, 92.52% zero-day detection, 0.0% PII leakage, and a BERTScore F1 of 0.8710, outperforming ReGAIN, Tri-LLM, and CyberRAG across all major evaluation dimensions. Through Maximal Marginal Relevance (MMR) search, the framework's Immunity Engine acts as a digital immune system, seamlessly translating zero-day threat patterns discovered at one node into actionable firewall configurations for automated deployment and proactive defense across the broader network.